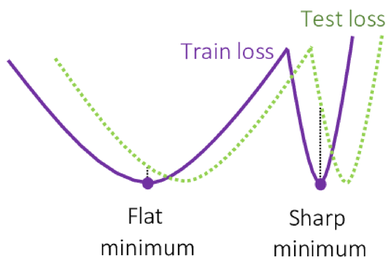
Sharpness-Aware Minimization: Feature Selection and Implicit Biases

Samuel Schapiro, Han Zhao

University of Illinois, Urbana-Champaign

Background

Sharpness-Aware Minimization (SAM) is an optimization algorithm that seeks flat minima for improved generalization.



It makes a simple modification to the gradient descent update rule at each step:

$$w_{t+1} \leftarrow w_t - \nabla_w L(w)\big|_{w_t + \rho \frac{\nabla_w L(w_t)}{\|\nabla_w L(w_t)\|}}, \quad \rho \in \mathbb{R}^+$$

SAM has been extremely successful empirically, reaching **state-of-the-art performance on SVHN, CIFAR-10**, and other benchmarks. SAM also **improves adversarial robustness, transfer learning, and fine-tuning** performance. However, it is now understood that flatness alone is not the only reason for SAM's improvement.

Contributions

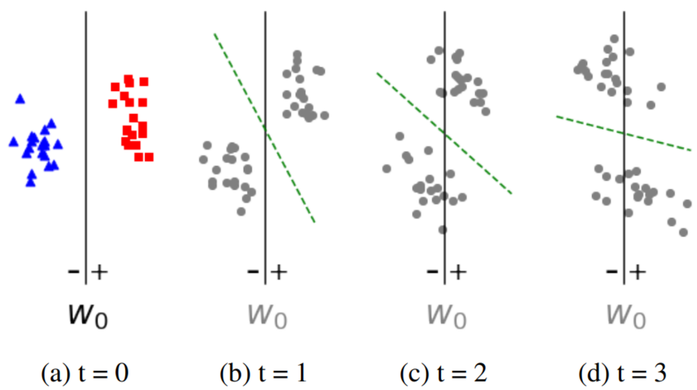
In light of SAM's successes and the mysteries surrounding its state-of-the-art performance, we study the following two questions:

- Can SAM improve performance for **gradual domain adaptation**? Does it do this by learning more **invariant representations**?
- How can we understand the **feature selection process** of SAM?

We find that SAM offers **significant improvements for gradual domain adaptation**, but learning more **invariant representations may not be the cause.** On a toy problem, we find that SAM attenuates loss oscillations due to opposing signals, suggesting it **avoids unstable features.**

Gradual Domain Adaptation

Gradual domain adaptation (GDA) has emerged as a successful technique to mitigate problems in domain adaptation arising from poor performance when the distribution shift between the source and target domain is too large. GDA constructs intermediate domains – for which the average distribution shift is small – between the source and target domain, leading to stronger performance than standard DA.



SAM Improves Gradual Domain Adaptation Performance

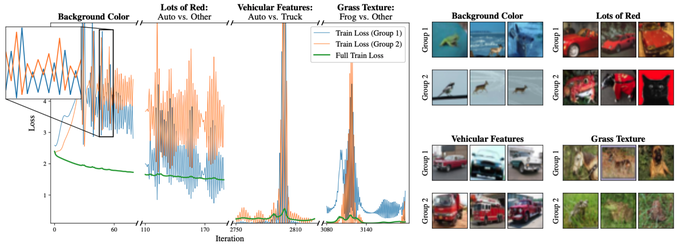
SAM offers an average improvement of 3.5% on all datasets. SAM offers an asymmetric payoff: it underperforms only on one dataset and by merely 0.4%, whereas it can improve accuracy up to 9.3%.

Dataset	Method	Target Acc %	Weight Shift
Color MNIST	SAM (T = 20)	**95.35**$_{\pm 0.1}$	0.70$_{\pm 0.01}$ (0.866)
	Adam (T = 10)	91.22$_{\pm 0.21}$	**0.68**$_{\pm 0.02}$ (0.961)
Rotated MNIST	SAM (T = 100)	**96.21**$_{\pm 0.705}$	1.03$_{\pm 0.01}$ (0.868)
	Adam (T = 100)	86.91$_{\pm 9.98}$	**0.96**$_{\pm 0.03}$ (0.991)
Covertype	SAM (T = 50)	**71.43**$_{\pm 2.25}$	**0.57**$_{\pm 0.04}$ (0.28)
	Adam (T = 10)	70.44$_{\pm 2.94}$	0.81$_{\pm 0.04}$ (0.19)
Portraits	SAM (T = 10)	82.85$_{\pm 3.75}$	**0.34**$_{\pm 0.01}$ (-0.16)
	Adam (T = 7)	**83.25**$_{\pm 2.57}$	0.352$_{\pm 0.02}$ (0.16)

However, SAM does not convincingly learn more invariant representations on all datasets.

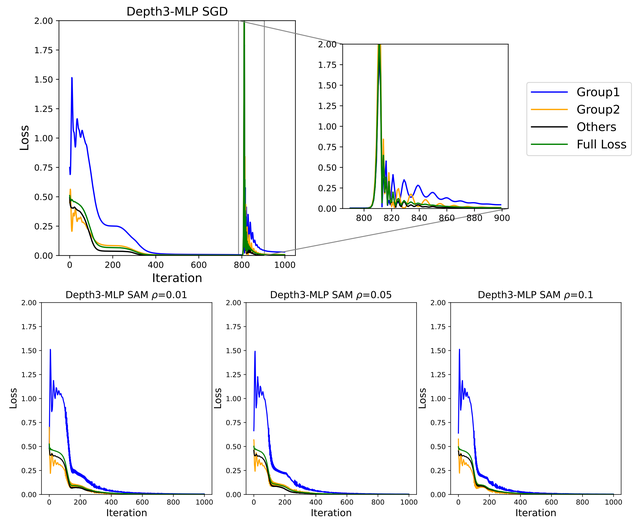
Feature Selection and Outliers with Opposing Signals

Outliers with opposing signals are consistent, large magnitude features that provide gradients pointing in opposite directions and dominate the network output throughout training. It is believed that they cause progressive sharpening of the Hessian spectral norm during training, leading to the edge-of-stability, and play a role in understanding the feature selection process of neural networks.



SAM Reduces Oscillations Due to Opposing Signals

On synthetic datasets constructed from Chebyshev polynomials, SAM dampens group-wise loss oscillations due to the opposing signal phenomenon, suggesting an implicit tendency to avoid statistically weak features that (i) induce opposing gradients on distinct groups of data points and (ii) are unlikely to generalize



sjs17@illinois.edu